

Mouli Chittabathina

Data Analyst & Data Engineer

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Professional Summary

Results-driven Data Analyst and Engineer with expertise in machine learning, data visualization, and cloud technologies. Proven track record of delivering high-impact data solutions that drive operational efficiency and business intelligence. Skilled in deriving actionable insights from complex datasets using advanced analytics techniques and developing robust data pipelines. Committed to continuous learning and applying cutting-edge technologies to solve challenging business problems.

Professional Experience

August 2024 – Present **Data Analyst, Indukuri Information Services, TN, USA (Remote)**

- Engineered SQL-based analytics solutions on Microsoft SQL Server, SSMS, and AWS, achieving 100% data accuracy improvement and a 25% boost via refined queries and structured pipelines.
- Enhanced ETL workflows with Apache Airflow, Python, PySpark, and SQL, increasing operational efficiency by 30% and reducing data latency by 20%.
- Leveraged Jupyter Notebook, Google BigQuery, Microsoft Fabric, and Snowflake to process 10M+ records; upgraded queries on SQL Server, PostgreSQL, MySQL, NoSQL, reducing processing time by 30%.
- Implemented data quality frameworks and automated validation processes, ensuring consistent and reliable data across enterprise systems.
- Collaborated with cross-functional teams to translate business requirements into technical specifications and data models.

January 2024 – May 2024 **Data Analyst, Hildreth Institute, MA, USA (Remote)**

- Created interactive dashboards and KPIs in Tableau, Power BI, QuickSight, and Looker, improving decision-making efficiency by 20% for 3,000+ institutions by visualizing key performance metrics and streamlining reporting workflows.
- Automated data extraction from APIs and web scraping using Python and SQL, cutting manual entry by 90% while ensuring seamless integration with AWS S3 and Redshift, enhancing storage and retrieval processes.
- Conducted exploratory data analysis (EDA) using pandas, NumPy, Excel VBA, and SQL, identifying trends that improved data models by 18% and maximized analytical accuracy by 15% through statistical techniques.
- Delivered presentations to senior leadership, effectively communicating complex data insights and recommendations.
- Mentored junior team members on advanced analytics techniques and best practices for data visualization.

April 2021 – December 2022 **Senior Software Engineer, DMI Finance, New Delhi, India**

- Optimized data pipelines using Snowflake, DynamoDB, Qlik, and Kafka, improving speeds by 40% and reducing costs by 25%.
- Synchronized workflows by integrating IBM Cloud, Azure Cloud, and GCP into CI/CD pipelines, leveraging automated containerization and Infrastructure as Code (IaC) to cut deployment time by 35% while enhancing scalability.
- Led a team of 5 engineers in developing and maintaining data-driven applications, fostering a collaborative environment that improved team productivity.
- Established data governance protocols and documentation standards to ensure compliance with industry regulations.
- Participated in architectural design decisions and technology selection for enterprise data solutions.

June 2020 – March 2021 **Software Engineer, DMI Finance, New Delhi, India**

- Configured monitoring systems via Linux scripting and SQL, increasing uptime by 20% and reducing oversight by 40%.
- Collaborated with teams to integrate cloud solutions, improving batch processing by 20% and optimizing system performance.
- Developed and maintained ETL processes for data integration across multiple business systems.
- Created comprehensive documentation for system architecture and operational procedures.
- Participated in Agile development ceremonies and contributed to continuous improvement initiatives.

Technical Skills

Programming Languages	Python, SQL, PySpark, R, Scala
Frameworks & Libraries	pandas, NumPy, matplotlib, seaborn, plotly, scikit-learn, tidyverse, dplyr, ggplot2, PyTorch, TensorFlow
Data Tools & Platforms	Tableau, Power BI, Qlik, QuickSight, Looker, SSRS, Excel, Jupyter Notebook, Azure Databricks, Microsoft Fabric, Kafka, Hadoop, Hive, AWS (Glue, S3, Lambda, Redshift), Google BigQuery, Snowflake, Salesforce CRM, SQL Server, PostgreSQL, MySQL, NoSQL
Development & Collaboration Tools	Azure Cloud, GCP, SQL Server, SSIS, Docker, Kubernetes, Git, Jenkins, Linux, CI/CD Pipelines
Data Concepts	Data Analysis, EDA, Business Analysis, Data Visualization, Machine Learning, NLP, Big Data, ETL, DAGs, Statistics, Hypothesis Testing, A/B Testing, Data Warehousing, Data Mining, Data Governance, Data Quality, Cloud Computing, DBMS
Additional Skills	Agile Methodology, Data Pipeline, Data Modeling, Data Lakes, Data Security, Data Integrity

Projects

- January 2025 **Fraud Detection Enhancement**, Technologies: Python, SQL, PySpark, scikit-learn, TensorFlow, Power BI, AWS/GCP
- Formulated machine learning models using scikit-learn and TensorFlow, improving fraud detection accuracy from 78% to 97%.
 - Built data pipelines in PySpark and SQL on AWS and GCP, processing 100M+ transactions and reducing latency by 40%.
 - Implemented real-time alerting system for suspicious transaction detection.
 - Created executive dashboards for monitoring key fraud metrics and trends.
 - Designed comprehensive documentation and knowledge transfer procedures for system maintenance.
- November 2024 **Sales Forecasting Dashboard**, Technologies: Time Series, Power BI, Tableau, Qlik, Looker, SQL, Python, Excel
- Devised a forecasting model with Python/SQL time series analysis and Excel Pivot Tables, improving accuracy from 65% to 95% and reducing inventory shortages by 20%.
 - Crafted 10+ interactive dashboards with KPIs in Power BI, Tableau, Qlik, and QuickSight, accelerating decision-making by 35%.
 - Developed automated ETL processes to ensure data consistency across visualization platforms.
 - Implemented mobile-friendly reporting capabilities for executive stakeholders.
 - Designed custom metrics to address specific business needs and enhance analytical capabilities.
- August 2024 **Real-Time Analytics Platform**, Technologies: Apache Kafka, Hadoop, PySpark, AWS, Microsoft Fabric, Google BigQuery
- Architected ingestion pipelines with Apache Kafka and Hadoop, processing 5M+ events daily and boosting speed by 35%.
 - Streamlined data transformation with PySpark, Microsoft Fabric, and Google BigQuery, cutting latency from 10s to 7s and yielding 20% cost savings.
 - Implemented comprehensive monitoring and alerting systems to ensure platform reliability.
 - Designed disaster recovery and high-availability strategies for critical components.
 - Collaborated with data scientists to integrate ML models into the real-time processing pipeline.
- May 2024 **Retail Segmentation & Analysis**, Technologies: SQL, R, Tableau, Power BI, QuickSight, Excel Array Functions
- Analyzed customer transactions with SQL, R, and Excel Power Pivot to segment high-value groups contributing 40% of total sales.
 - Executed clustering algorithms and machine learning techniques, improving targeted marketing effectiveness by 25%.
 - Developed automatic refresh mechanisms to incorporate new customer data into segmentation models.
 - Created interactive visualizations allowing marketing teams to explore segment characteristics.
 - Designed and implemented a recommendation engine based on segment preferences and behavior patterns.

Education

- January 2023 – May 2024 **Master's in Data Analytics Engineering**, *George Mason University*, Fairfax, VA, GPA: 4.0/4.0
Relevant Coursework: Advanced Machine Learning, Big Data Analytics, Statistical Modeling, Database Management Systems, Data Mining, Predictive Analytics, Cloud Computing for Data Science, Natural Language Processing
- July 2016 – May 2020 **Bachelor's in Electronics and Communication Engineering**, *National Institute of Technology*, Jaipur, India, GPA: 7.86/10
Relevant Coursework: Programming Fundamentals, Data Structures and Algorithms, Signals and Systems, Digital Signal Processing, Computer Networks, Embedded Systems

Awards and Certifications

- Big Data with Spark and Hadoop | IBM
- SQL | 365 Data Science
- Excel | 365 Data Science
- Python | 365 Data Science
- Statistics | 365 Data Science
- Honor Society Member | Phi Kappa Phi

Professional Development

- Regular participation in data science meetups and conferences
- Active contributor to open-source data projects
- Continuous learning through online platforms (Coursera, Udemy, DataCamp)
- Member of professional data science communities and forums

Languages

- English (Professional Proficiency)
- Hindi (Native)
- Telugu (Native)

References

Available upon request